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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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1 STEP 1 – Exploratory Data Analysis

Training period: 5 Jan 2007 to 31 December 2014

Validation period: 1 Jan 2015 to 29 December 2017

Test period: 1 Jan 2018 to 30 December 2022

1.1 Individual distributions

ETF — asset class	Ann. ret %	Ann. vol %	Skew	Exc. kurt.	Worst day %	Best day %	Total ret %
SPY (Equity)	8.23	20.51	-0.36	13.12	-11.59	13.56	272.04
TLT (Fixed income)	3.62	15.21	0.02	3.64	-6.90	7.25	78.36
SHY (Cash-like)	1.43	1.40	0.00	7.68	-0.66	0.71	25.76
GLD (Precious metals)	6.49	17.72	-0.25	6.58	-9.19	10.70	181.93
DBO (Crude oil)	-2.41	32.08	-0.57	4.65	-18.21	10.10	-31.92

Table 1. Daily log-return statistics over the entire period of time from 5 Jan 2007 to 30 December 2022

We can see that crude oil is the most volatile asset with annualised volatility of 32.1%, and it has negative annual return over the 16 years of data. Equity has the largest excess kurtosis, meaning the fattest tail of 13.12. Figure 1 shows the distribution of the log-returns on a log-scale and the fat tail becomes apparent.

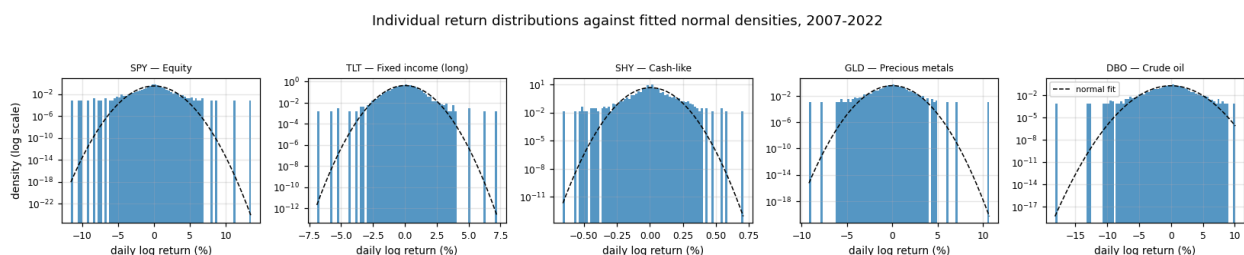


Figure 1. Distribution of daily return fitted with normal distribution. We see that extreme positive or negative returns happen more frequently than predicted by normal distribution i.e. the returns distribution are fat-tailed.

If we invest \$100 at the start of 2007 into each of the five asset classes and held them for the entire duration until the end of 2022, Figure 2 shows the growth of the \$100. We also plot the rolling 21-day annualised volatility, and we observe volatility clustering.

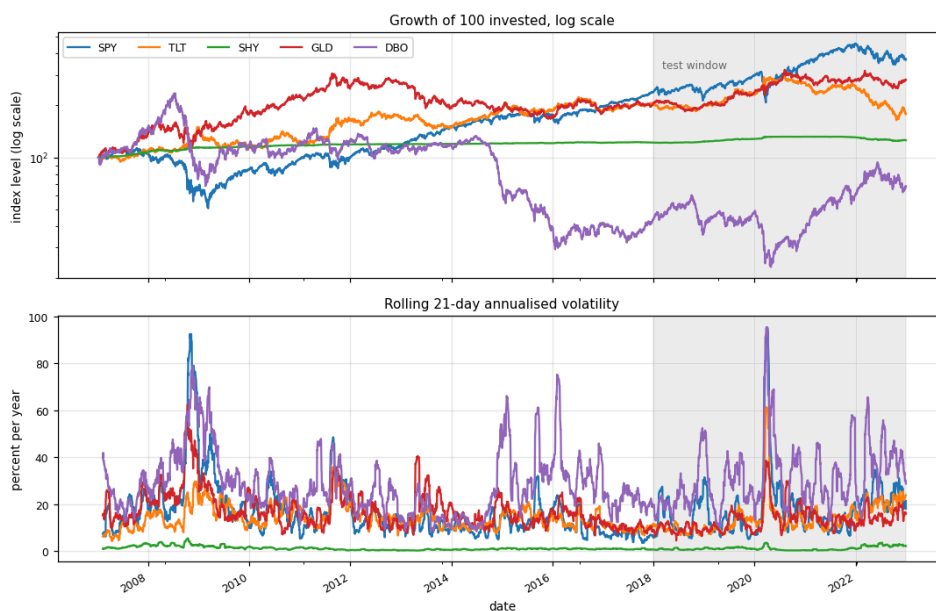


Figure 2. Cumulative returns of 100 invested at the beginning plotted on a log scale (upper) and rolling 21-day annualised volatility (lower). The shaded band is the test window.

1.2 Joint Distribution

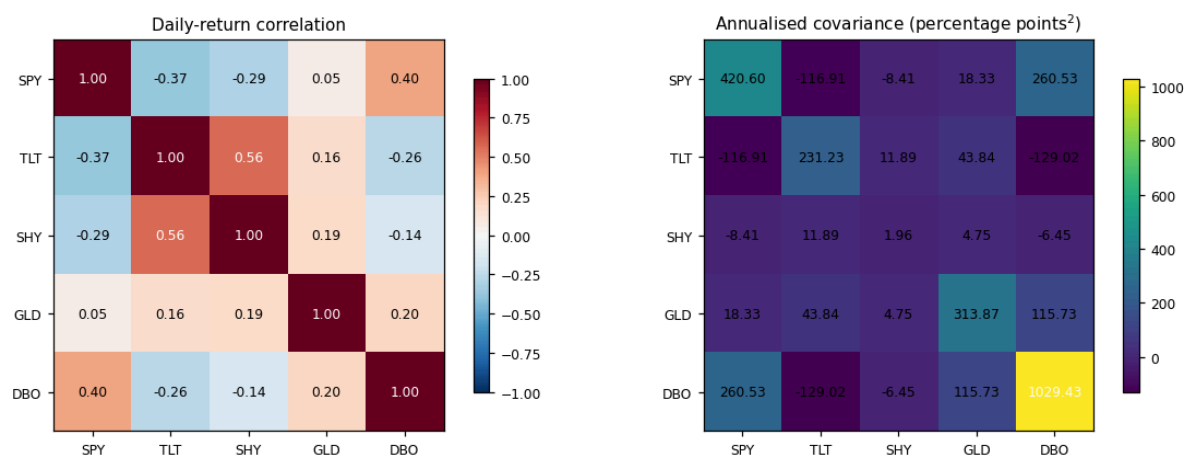


Table 3. Correlation (left) and covariance (right) matrix of daily log returns

A few observations:

1. Equity and long treasuries are negatively correlated at -0.37
2. TLT and SHY, which are the two treasury fund are relatively correlated at 0.57.
3. Gold is the least correlated with the rest of the assets, making it the best diversification tool, with the largest correlation being with DBO crude oil at 0.2.

1.3 Stationarity and the representation fed to the models

ETF	Log price (constant)	Log price (+ trend)	Daily log return	25-day log return
SPY	-0.03	-3.05	-15.46	-8.02
TLT	-1.94	-1.35	-12.17	-7.68
SHY	-4.37	-3.12	-9.66	-5.34
GLD	-2.59	-2.59	-63.77	-7.90
DBO	-1.23	-1.61	-66.53	-6.47

Table 4. Augmented Dickey-Fuller statistics, lag order chosen by AIC. MacKinnon 5% critical values are -2.86 with a constant and -3.41 with a constant and trend;

As we can see from the Augmented Dickey-Fuller test results, if we included trend then we cannot reject the null hypothesis for all the five assets, meaning they are non-stationary. However, if we use daily log return, the unit root can be rejected and the data is stationary. Hence, we will use log daily returns. Daily log returns are employed as these offer a better stationarity representation of the price series and are also additive over time. Daily log returns are further justified by the ADF test as the null hypothesis that a unit root exists is rejected for all five ETFs under study. Daily log returns make a good choice for constructing the 25-day cumulative return target.

2 STEP 2 – MLP

We use a lookback window size of 30, this forms the input vector. We then pass it through the 30x32 ReLU followed by a dropout rate of 0.2. We pass it to another 32x16 ReLU and then finally the output layer to output the 25-day ahead log return of the asset./

In-sample coefficient of determination:

SPY	TLT	SHY	GLD	DBO	Mean
+0.136	+0.150	+0.403	+0.094	+0.110	+0.179

Out-of-sample coefficient of determination:

SPY	TLT	SHY	GLD	DBO	Mean
-0.052	+0.013	+0.050	-0.007	-0.037	-0.007

Out-of-sample information coefficient:

SPY	TLT	SHY	GLD	DBO	Mean
-0.149	+0.067	+0.081	+0.026	+0.024	+0.010

Comment: Out-of-sample coefficient of determination is much lower than in-sample and negative as well, suggesting MLP struggles to find patterns in the noisy financial data. The substantial deterioration from in-sample to out-of-sample R^2 is consistent with overfitting and/or a weak and unstable predictive signal.

3 STEP 3 – CNN-GAF

In-sample coefficient of determination:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	+0.136	+0.150	+0.403	+0.094	+0.110	+0.179
CNN-GAF	+0.037	+0.042	+0.166	+0.072	+0.058	+0.075

Out-of-sample coefficient of determination:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	-0.052	+0.013	+0.050	-0.007	-0.037	-0.007
CNN-GAF	+0.001	+0.010	+0.049	-0.038	-0.011	+0.002

Out-of-sample information coefficient:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	-0.149	+0.067	+0.081	+0.026	+0.024	+0.010
CNN-GAF	+0.032	+0.008	+0.071	-0.114	-0.003	-0.001

Comment: The out-of-sample performance of CNN-DAF is not as bad as MLP, though we can see that it also does not fit the in-sample data well from the low coefficient of determination score. This is a classic example of bias-variance trade-off of low variance and high bias. This is because when we represent the data as a 30x30 image, whereby each pixel is a pair-wise angular relationship between two days. This makes it good at capturing local relationships between data such as intense consecutive price movements because of volatility clustering.

4 STEP 4 – LSTM

In-sample coefficient of determination:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	+0.136	+0.150	+0.403	+0.094	+0.110	+0.179
CNN-GAF	+0.037	+0.042	+0.166	+0.072	+0.058	+0.075
LSTM (single-output)	+0.206	+0.137	+0.315	+0.091	+0.112	+0.172

Out-of-sample coefficient of determination:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	-0.052	+0.013	+0.050	-0.007	-0.037	-0.007
CNN-GAF	+0.001	+0.010	+0.049	-0.038	-0.011	+0.002
LSTM (single-output)	-0.188	+0.017	-0.181	-0.013	+0.036	-0.066

Out-of-sample information coefficient:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	-0.149	+0.067	+0.081	+0.026	+0.024	+0.010
CNN-GAF	+0.032	+0.008	+0.071	-0.114	-0.003	-0.001
LSTM (single-output)	-0.193	+0.082	-0.132	+0.061	+0.167	-0.003

Comment: Mean in-sample performance of LSTM is similar to MLP at .172 vs 0.179. Another thing we noticed is that all three models: MLP, CNN and LSTM seem to fit SHY the best, which might be a result of its low volatility feature. LSTM suffer the same overfitting problem as MLP with its out-of-sample degrading considerably. It has the worst out-of-sample coefficient of determination of -0.066 among MLP, CNN and LSTM. It seems that the best architecture for SPY is CNN, TLT is LSTM, SHY is MLP, DBO is LSTM. All three architectures perform badly on GLD. In general, GLD and DBO fits the worst because their returns are dominated by shocks and they are heavily affected by news. On the hand, TLT offers the best fit among all models because their returns are generally smoother since they are affected mainly by treasury yield which moves slower. This is why LSTM works well for TLT because treasury yield might persist at a similar level over many years and the LSTM would be able to learn the past history pretty well.

5 STEP 5 – Multi-output Model

In-sample coefficient of determination:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	+0.136	+0.150	+0.403	+0.094	+0.110	+0.179
CNN-GAF	+0.037	+0.042	+0.166	+0.072	+0.058	+0.075
LSTM (single-output)	+0.206	+0.137	+0.315	+0.091	+0.112	+0.172
LSTM (multi-output)	+0.324	+0.300	+0.438	+0.241	+0.371	+0.335

Out-of-sample coefficient of determination:

Architecture	SPY	TLT	SHY	GLD	DBO	Mean
MLP	-0.149	+0.067	+0.081	+0.026	+0.024	+0.010
CNN-GAF	+0.032	+0.008	+0.071	-0.114	-0.003	-0.001
LSTM (single-output)	-0.193	+0.082	-0.132	+0.061	+0.167	-0.003
LSTM (multi-output)	+0.176	+0.054	+0.019	+0.017	+0.195	+0.092

Forecast RMSE in test sample of the five separate LSTMs against the single multi-output LSTM.

ETF — asset class	Single-output RMSE %	Multi-output RMSE %	Change	Single-output R^2	Multi-output R^2
SPY — Equity	6.72	6.07	-9.6%	-0.188	+0.029
TLT — Fixed income (long)	4.78	4.80	+0.5%	+0.017	+0.007
SHY — Cash-like	0.59	0.53	-9.1%	-0.181	+0.024
GLD — Precious metals	4.07	4.31	+6.0%	-0.013	-0.138
DBO — Crude oil	10.81	10.77	-0.4%	+0.036	+0.044

*A negative change means the multi-output model forecasts that asset more accurately.

Comment:

- Clearly, the multi-output model fits the in-sample data the best among all because it uses the most information from the 5 LSTM models, with a mean of 0.335 coefficient of determination.
- The multi-output model improves the equity forecast by 9.6% of RMSE and the short-Treasury forecast by 9.1%. However, it forecasted gold worse than a single-output LSTM, by 6.0% of RMSE and from -0.013 to -0.138.
- The underperformance of gold is likely due to its low correlation with the other asset classes.
- The advantage of the multi-output model is that when there is a common predictor among all the three assets, making all of them pass through a single recurrent state we get some regularisation effect. At the same time, we also get to work with more data. This helps the model to filter out asset-specific noise and generalise better.
- The two architectures are not using the same information. The multi-output model can see cross-asset relationships while the single-output one cannot. This is evident from the fact that the multi-output model is also the only architecture whose mean information coefficient is positive.

6 STEP 6 – Trading Strategy & Backtest

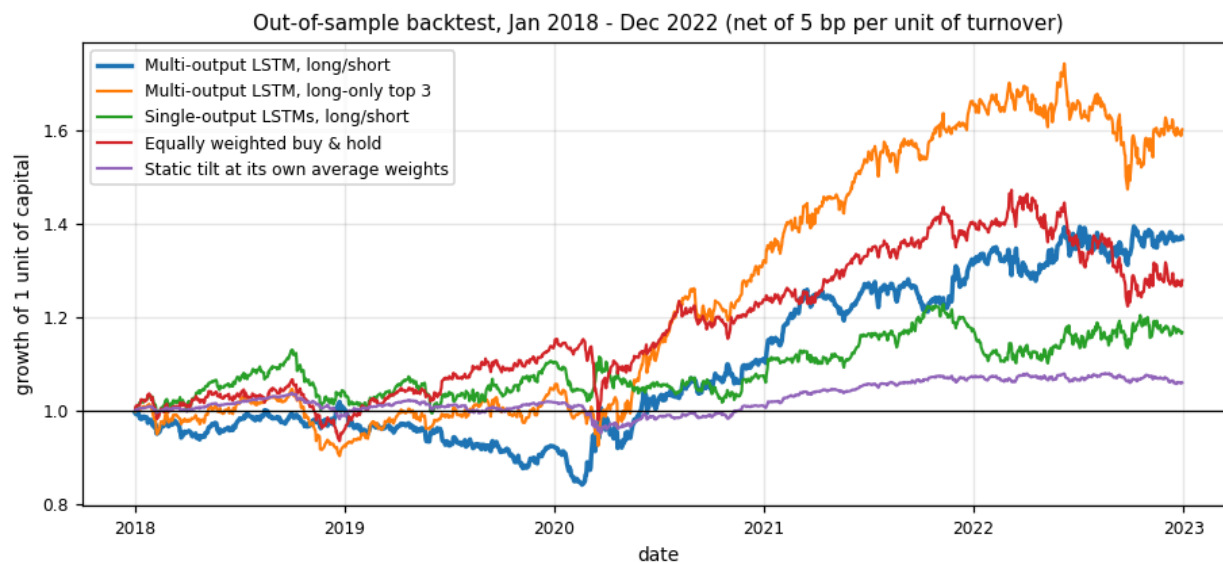
We devised three trading strategies and two reference portfolios to be compared over the entire test window.

- **(Strategy 1) Multi-output long/short:** long the top two ranked asset classes and short the bottom two, allocating 25% of capital each. Dollar-neutral, gross exposure 1.0.
- **(Strategy 2) Multi-output long-only** — equal weight on the top three.
- **(Strategy 3) Single-output long/short** — identical rules driven by the five separate Step-4 models, which separates the value of the multi-output architecture from the value of the recurrent cell itself.
- **(Reference 1) Equally weighted buy and hold** — the benchmark, bought at the first test close and held throughout.
- **(Reference 2) Long/short strategy's average weights, buy and hold.**

The results for the out-of-sample backtest from 1 Jan 2018 to 30 December 2022 are summarised in the table below:

Strategy	Total ret %	CAGR %	Ann. vol %	Sharpe	Max DD %	Rebal.	Turnover
Multi-output LSTM, long/short	36.82	6.48	11.47	0.56	-17.40	50	52.5
Multi-output LSTM, long-only top 3	60.13	9.88	11.99	0.82	-15.44	50	35.7
Single-output LSTMs, long/short	16.68	3.14	9.68	0.32	-12.00	50	40.5
Equally weighted buy & hold	27.86	5.04	10.20	0.49	-16.90	0	0.0
Long/short strategy's average weights, buy and hold	5.96	1.16	3.45	0.34	-8.29	0	0.0

The multi-output long only strategy gave the best backtest performance among all the strategies and reference portfolio, returning 60.13% over the test period, and has a Sharpe ratio of 0.82. It beats the equal weight buy-and-hold benchmark in terms of returns and drawdown. The strategy has a drawdown of -15.44% compared to -16.9% of the reference portfolio.



Cumulative returns graph of the three strategies and the two reference portfolio.

7 STEP 7 – Discussion

7.1 Comparative predictability

MLP is the most restrictive model so it is the least able to encode any regime information. CNN filters for local behaviour which make them good at capturing returns when volatility clusters. LSTM are able to learn the past price trajectory well due to its memory property. Hence, they tend to fare well for assets which are low in volatile. However, this also mean it is prone to overfitting because it would be hard for it to respond quickly to any change to macroeconomics over time.

7.2 Multi-output v.s. single-output

They are not capturing the same information.

- The single-output models cannot capture any cross-asset relationship;
- The multi-output model is also the only architecture with a positive mean information coefficient (+0.092).

7.3 Backtest performance

Compared to an equally weighted buy-and-hold benchmark returning 27.86%, the multi-output long-only strategy returned 60.13% while the long/short variation returned 36.82%, both with a higher Sharpe ratios.

7.4 Limitations of the project

- **Small effective sample size:** with just 1,957 training samples for machine learning models with thousands of parameters, and in a domain with a very low signal-to-noise ratio. It is hard to completely trust these results.
- **Only a single test period:** a walk-forward validation test across several windows provide a much better test of the strategy's performance.
- **Constant transactional costs:** backtest assumes five basis points per unit of turnover, no borrowing cost.
- **Low dimensional inputs:** only price data is used to train the model. In reality, other forms of data like macroeconomic releases, the shape of the yield curve, implied volatility and cross-asset flow data are used.